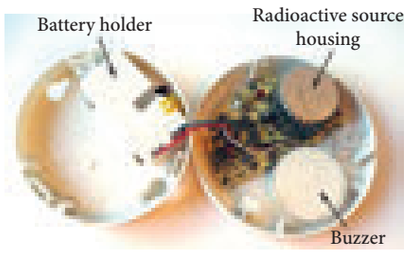
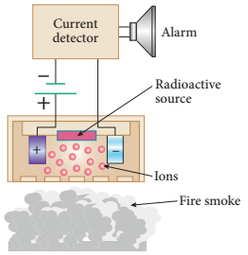


A very interesting application of alpha decay is in smoke detectors which prevent us from any hazardous fire.







The smoke detector uses around 0.2 *mg* of man-made weak radioactive isotope called americium ($^{241}_{95}Am$). This radioactive source is placed between two oppositely charged metal plates and α radiations from $^{241}_{95}Am$ continuously ionize the nitrogen, oxygen molecules in the air space between the plates. As a result, there will be a continuous flow of small steady current in the circuit. If smoke enters, the radiation is being absorbed by the smoke particles rather than air molecules. As a result, the ionization and along with it the current is reduced. This drop in current is detected by the circuit and alarm starts.

The radiation dosage emitted by americium is very much less than safe level, so it can be considered harmless.

neutron (not inside any nucleus) can exhibit β^- decay.

It is important to note that the electron or positron which comes out from nuclei during beta decay are not present inside the nuclei but they are produced only during the conversion of neutron into proton or proton into neutron inside the nucleus.

Neutrino:

Initially, it was thought that during beta decay, a neutron in the parent nucleus is converted into the daughter nuclei by emitting only electron as given by

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^- \qquad (9.30) $$

But the kinetic energy of electron coming out of the nucleus did not match with the experimental results. In alpha decay, the alpha particle takes only certain allowed discrete energies whereas in beta decay, it was found that the beta particle (i.e, electron) has a continuous range of energies. But the conservation of energy and momentum gives specific single values for energy of electron and the recoiling nucleus Y. It seems that the conservation of energy, momentum are violated and could not be explained why energy of beta particle having continuous range of values. So beta decay remained as a puzzle for several years.

After a detailed theoretical and experimental study in 1931, W.Pauli proposed a third particle which must be emitted in the beta decay process carrying away missing energy and momentum. Fermi later named this particle as *neutrino* (little neutral one) since it mass is small and is neutral carrying no charge. For many years, the neutrino (symbol ν, Greek *nu*) was hypothetical and could not be verified experimentally. Finally, the neutrino was

UNIT-9(XII-Physics_Vol-2)_New.indd 172 21-07-2020 15:49:57